

NRES 746 – Workout of the Day #1: Math & Stats Self-Diagnostic

Name: _____

This is **not graded**, but will be collected at the end of class. There are no wrong answers – the goal is just to take stock of the tools you're bringing into this class, and to see which ones might need a bit of dusting off. Some of these are meant to make you think a little!

Instructions: Work alone for about 5 minutes. Don't worry if you get stuck – just note where you got stuck and move on. After 5 minutes, compare answers with a neighbor for a few minutes.

Part 1: Logarithms & exponents

1a. Solve for x : $2^x = 32$

$x =$ _____

1b. A population grows according to $N_t = N_0 e^{rt}$. If $N_0 = 100$ and $N_t = 550$ after $t = 8$ years, solve for the growth rate r :

$r =$ _____

Part 2: Derivatives & integrals

2a. If $f(x) = (4x^2 - 3x)^5$, what is $f'(x)$?

$f'(x) =$ _____

2b. Evaluate $f'(x)$ from 2a at $x = 1$:

$f'(1) =$ _____

In one sentence, what does this number tell you about the function $f(x)$ at $x = 1$?

2c. A recruitment rate for a population is given by $r(t) = 4t$ (new individuals per unit time).

First, find the antiderivative $R(t) = \int 4t \, dt$:

$R(t) =$ _____

Now use $R(t)$ to compute the total number of new recruits between $t = 0$ and $t = 5$:

$\int_0^5 4t \, dt = R(5) - R(0) =$ _____

Part 3: Probability

3a. $P(A) = 0.4$, $P(B) = 0.5$, and $P(A \text{ and } B) = 0.2$. What is $P(A \text{ or } B)$?

3b. A family has two children. You are told that at least one of them is a girl. What is the probability that *both* children are girls? Briefly show your reasoning (consider all the equally likely possibilities for two children).

Part 4: Confidence intervals & p-values

4a. Circle the interpretation you think is *most correct* for a 95% confidence interval on a mean:

- a) There is a ~95% probability that the true mean falls within this particular interval.
- b) About 95% of the intervals we compute from random samples would contain the true mean.
- c) 95% of the data fall within this interval.

4b. Circle the definition you think is *most correct* for a p-value:

- a) The probability that the null hypothesis is true.
- b) The probability of observing results at least as 'extreme' under the null hypothesis.
- c) The probability that the alternative hypothesis is false.

Part 5: R / pseudocode

5. Each snippet below has exactly one error. Circle it and write your correction directly on the snippet.

5a. Bracket notation

```
site_data <- matrix(c(12, 7, 19, 4, 45, 23, 61, 15), nrow = 4, ncol = 2)
# Extract the entire second column
site_data[2]
```

5b. For loops

```
values <- c(10, 20, 30, 40)
doubled <- numeric(length(values))
for (i in values) {
  doubled[i] <- values[i] * 2
}
```

5c. Function syntax

```
GetRange <- function(x)
  max_val <- max(x)
  min_val <- min(x)
  max_val - min_val
```

Self-rating

For each part above, circle how confident you felt (1 = totally lost, 5 = very confident):

Part	1	2	3	4	5
Logs & exponents	☐	☐	☐	☐	☐
Derivatives/integrals	☐	☐	☐	☐	☐
Probability	☐	☐	☐	☐	☐
Confidence intervals/p-values	☐	☐	☐	☐	☐
R / pseudocode	☐	☐	☐	☐	☐